MAPPING THE REPRESENTATIVENESS GAP: A PROBABILISTIC FRAMEWORK FOR URBAN AIR QUALITY AND POPULATION EXPOSURE

MIGUEL ROSA [1], ANDREU JULIAN-IZQUIERDO [2,3], CRISTINA CARNERERO [2], JAN ARMENGOL [2,3], VERA RODRIGUES [1], DIOGO LOPES [1]

[1] CESAM, Department of Environment and Planning, University of Aveiro, Portugal
[2] BARCELONA SUPERCOMPUTING CENTER, Barcelona, Spain
[3] DEPARTMENT OF FLUID MECHANICS, Universitat Politècnica de Catalunya, Barcelona, Spain

INTRODUCTION

⟶ **Regulatory Context**: The revised Ambient Air Quality Directive (Directive (EU) 2024/2881) establishes the Spatial Representativeness Area (SRA) as a critical metric for network design, compliance, and population exposure assessment.

⟶ **The Traditional Limitation**: Standard SRA approaches rely primarily on deterministic models and fixed similarity thresholds. These methods smooth microscale concentration gradients and ignore spatial uncertainty.

⟶ **The Exposure Blind Spot**: In dense street canyons, point measurements from monitoring stations fail to capture steep local gradients, leading to systematic miscalculations in population and commuter exposure.

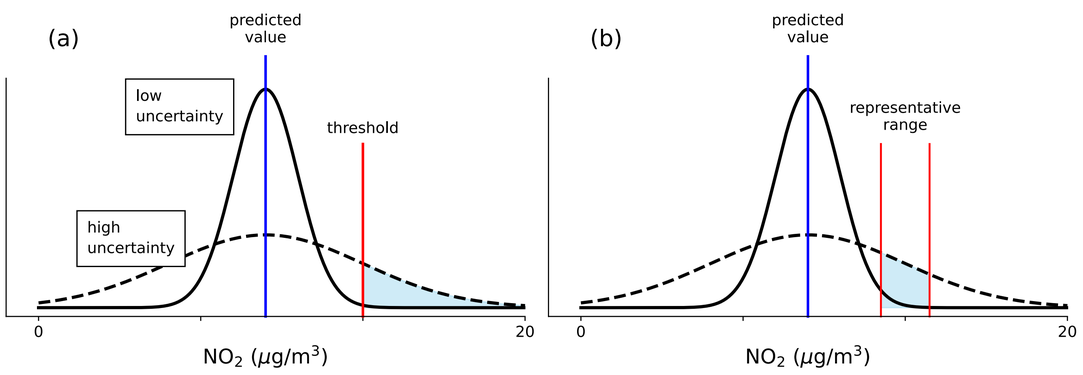
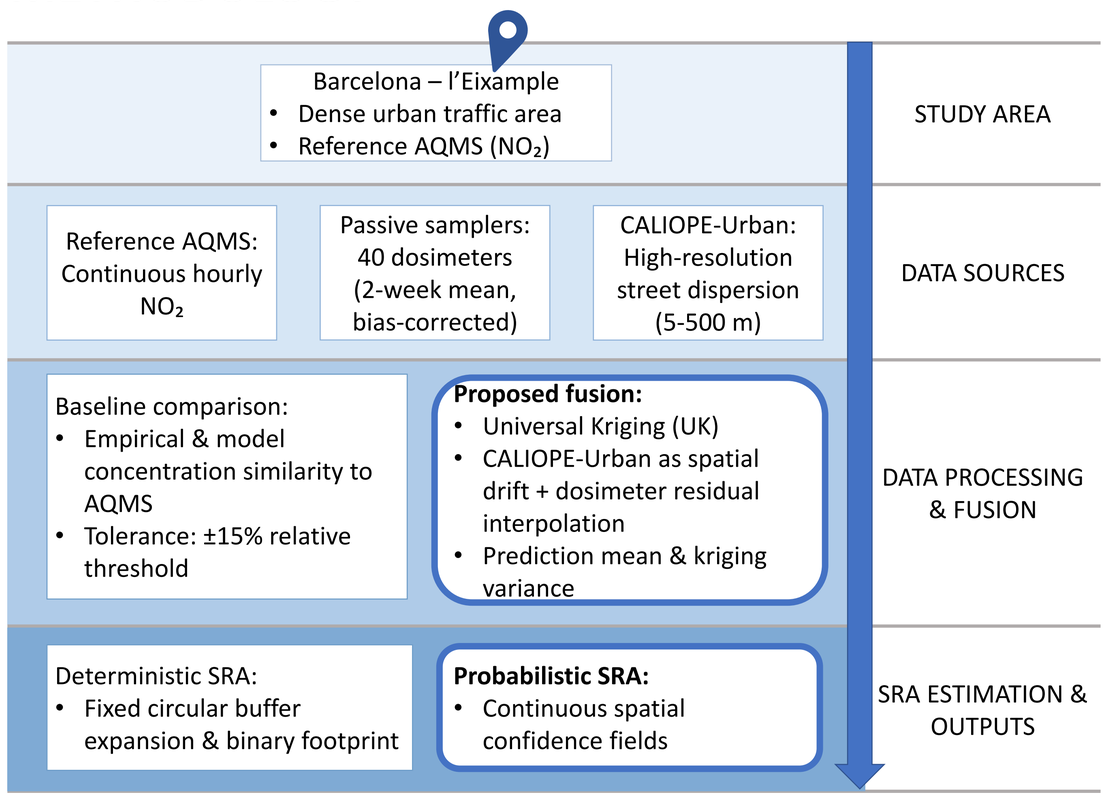


Figure 1. Schematic illustration of the probabilistic spatial representativeness framework: (a) threshold-based approach; (b) tolerance-interval approach. Shaded areas represent the probability associated with each framework.

METHODOLOGY



OBJECTIVE

To develop a high-resolution, uncertainty-aware method to quantify the Spatial Representativeness Area of urban air quality monitoring stations using data fusion techniques

RESULTS

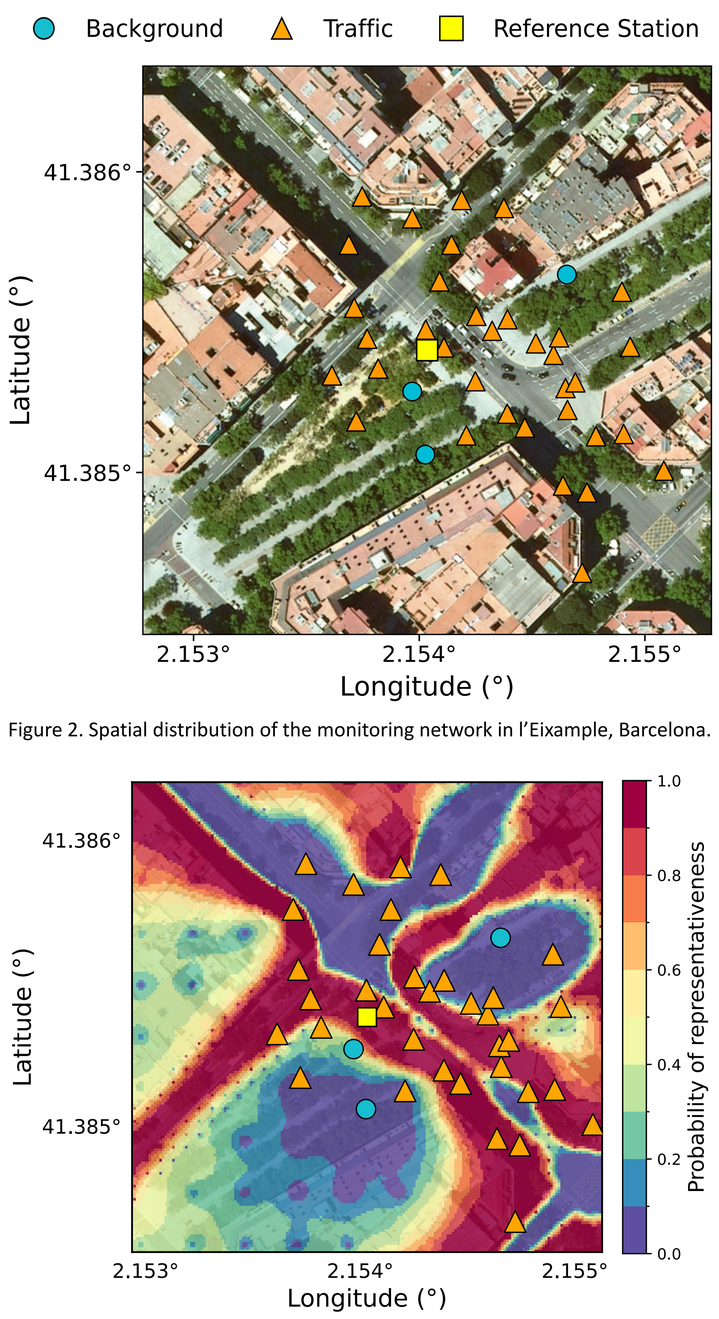


Figure 2. Spatial distribution of the monitoring network in l'Eixample, Barcelona.

Figure 3. Probability map of NO_2 spatial representativeness from Universal Kriging data fusion, showing the likelihood that concentrations fall within ±15% of the AQMS value.

- **Empirical SRA**: Restrictive buffers ranging from 20 m (passive dosimeters) to 24 m (model) and 26 m (Kriging);
- **Microscale gaps**: Point measurements fail to capture steep local concentration gradients;
- **High confidence (P > 0.75)**: Concentrated strictly along road edges and traffic-influenced corridors;
- **Spatial barriers**: Courtyards and buildings sharply drop probabilities, highlighting strong urban heterogeneity.

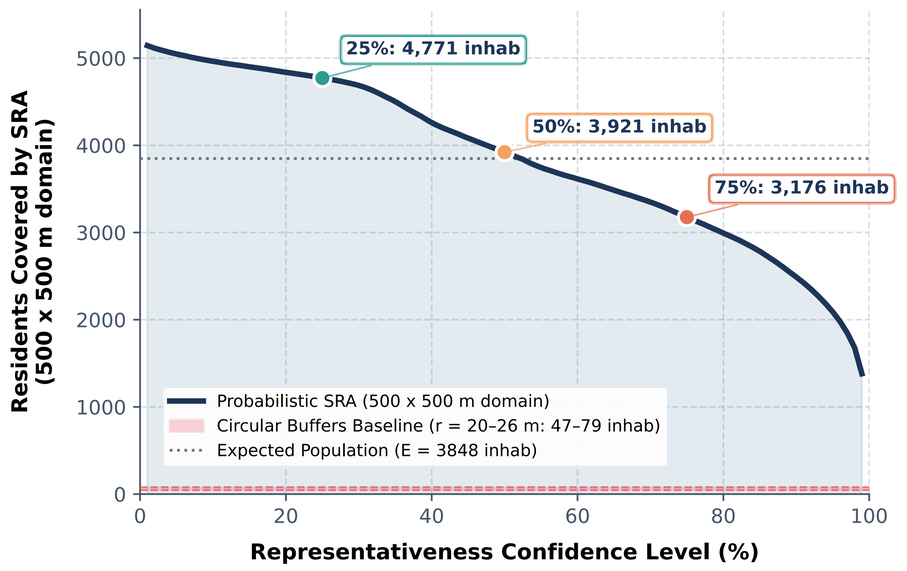


Figure 4. Sensitivity curve illustrating the expected number of residents covered by the SRA within the 500 x 500 m domain as a function of the representativeness confidence level, compared with the traditional circular buffer baseline (r = 20–26 m).

- **Domain constraint:** SRA counts restricted strictly to the 500 x 500 m study domain;
- **Buffer underestimation**: Traditional circular buffers (r = 20–26 m) capture an area of 0.13–0.21 ha, corresponding to only 47–79 residents;
- **Expected population**: Probabilistic data fusion yields an expected population of 3849 residents (3177–4773 inhabitants at 75%–25% confidence levels.

CONCLUSIONS

⟶ An **uncertainty-aware framework** combining passive dosimeters, street-scale dispersion modeling, and data fusion to quantify the **spatial representativeness** in urban environments;
⟶ **Microscale restrictions:** dense street-canyon environments show highly restricted SRAs (r = 20–26 m), highlighting steep local NO_2 gradients and strong spatial heterogeneity;
⟶ **Traditional circular buffers** prove overly conservative, capturing only between **47–79 residents** due to local spatial variations;
⟶ **Population coverage:** binary and probability-based SRA maps within the **500 x 500 m domain** yield an unexpected population coverage of **3849 residents**;
⟶ Explicit uncertainty quantification provides a robust interpretation of monitoring data, improving resident coverage assessment and regulatory compliance under the revised **EU Air Quality Directive**.

ACKNOWLEDGEMENTS

This work is funded by national funds through the FCT – Fundação para a Ciência e Tecnologia I.P., under the CESAM-Centro de Estudos do Ambiente e do Mar projects, references UID/50017/2025 (doi.org/10.54499/UID/50017/2025) and LA/P/0094/2020 (doi.org/10.54499/LA/P/0094/2020). Thanks are due to the SWIFT-AIR project (COMPETE2030-FEDER-00851800, Ref. FCT: 17119; DOI:10.54499/2023.18204.ICDT). Cristina Carnerero acknowledges her AI4S fellowship within the "Generación D" initiative by Red.es, Ministerio para la Transformación Digital y de la Función Pública, for talent attraction (C005/24-EDCV1), funded by NextGenerationEU through PRTR.